EXPERIMENT: Prolog Program for Student, Teacher and Subject

Aim

To write and implement a Prolog program to represent the relationship between students, teachers, and subjects using facts and rules.

Objective

- To understand facts and rules in Prolog.
- To represent student, teacher, and subject information.
- To find which teacher teaches a particular subject.
- To find which students study a particular subject.

Algorithm

5. Start the program.
6. Define facts for students.
7. Define facts for teachers.
8. Define facts for subjects.
9. Define relationships between teachers and subjects.
10. Define relationships between students and subjects.
11. Create rules to find the required relationships.
12. Execute queries to retrieve information.
13. Display the result.
14. Stop the program.

Flowchart

START

Define Student Facts

Define Teacher Facts

Define Subject Facts

Define Relationships

Enter Query

```
        |
        ▼
┌────────────────────────┐
│ Match Facts and Rules │
└───────────┬────────────┘          |
        ▼
┌────────────────────────┐
│ Display the Result   │
└───────────┬────────────┘
        |
        ▼
  ┌────────────────┐
  │    END    │
  └────────────────┘
```

Code

```
students = {
    "Arshad": ["AI", "Python", "Database"],
    "Rahul": ["Python", "Mathematics"],
    "Priya": ["AI", "Database"]
}
teachers = {
    "Dr_Nagarajan Babu": ["AI"],
    "Mr_Ravi": ["Python"],
    "Ms_Sneha": ["Database"],
    "Dr_Kumar": ["Mathematics"]
```

```
}
print("----- STUDENT DETAILS -----")
for student, subjects in students.items():
    print("Student:", student)
    print("Subjects:", ", ".join(subjects))
    print()
print("----- TEACHER DETAILS -----")
for teacher, subjects in teachers.items():
    print("Teacher:", teacher)
    print("Subjects:", ", ".join(subjects))
    print()
subject = input("Enter a subject to find its teacher: ")
found = False
for teacher, subjects in teachers.items():
    if subject in subjects:
        print("Teacher for", subject, "is", teacher)
        found = True
if not found:
    print("No teacher found for", subject)
student = input("Enter student name: ")
if student in students:
    print("Subjects studied by", student, "are:")
    for subject in students[student]:
        print("-", subject)
else:
```

print("Student not found")

Output

```
IDLE Shell 3.12.5
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.5 (tags/v3.12.5:ff3bc82, Aug  6 2024, 20:45:27) [MSC v.1940 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/shaik arshad basha/AppData/Local/Programs/Python/Python312/c
ode.py
----- STUDENT DETAILS -----
Student: Arshad
Subjects: AI, Python, Database

Student: Rahul
Subjects: Python, Mathematics

Student: Priya
Subjects: AI, Database

----- TEACHER DETAILS -----
Teacher: Dr_Nagarajan Babu
Subjects: AI

Teacher: Mr_Ravi
Subjects: Python

Teacher: Ms_Sneha
Subjects: Database

Teacher: Dr_Kumar
Subjects: Mathematics

Enter a subject to find its teacher: AI
Teacher for AI is Dr_Nagarajan Babu
Enter student name: Arshad
Subjects studied by Arshad are:
- AI
- Python
- Database
>>>
                                                                  Ln: 35  Col: 0
```

Result

The Prolog program was successfully implemented to represent and retrieve relationships between students, teachers, and subjects.

Conclusion

The program demonstrates the use of facts, rules, variables, and queries in Prolog. It can determine which subjects a student studies and which teacher teaches those subjects.